

HANRAN MA

PRACTICE-BASED MEDIA ARTIST & RESEARCHER

Game Engines / Interactive Media / Audiovisual Systems / Higher Education
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RESEARCH PROFILE

Practice-based media artist and researcher working with game engines, interactive systems, real-time audiovisual media, and networked or participatory art. My current PhD research examines game engines as artistic infrastructures: environments in which state logic, event relations, audiovisual interfaces, and bounded generative processes shape narrative, perception, and audience agency. Across Unity-based works, creative coding, sound-responsive systems, and installations, I treat technical structures not as invisible production tools but as materials that can organise artistic experience and generate research knowledge.

RESEARCH INTERESTS

Practice-based Artistic Research	Game Engines as Artistic Infrastructures	State-Based Narrative	Audiovisual Interface Orchestration
Generative Media Systems	Interactive & Networked Art	Embodied Interaction & Accessibility	Software / Interface Culture

EDUCATION

M.A. in Interactive Media Arts (IMA)

New York University | 2021-2022

B.A. in New Technologies of Art

Accademia Albertina di Belle Arti di Torino, Turin, Italy | 2014-2017

PUBLICATION

Ma, H. (2026). *Design and Implementation of a Unity-Based Experimental Walking Simulation with Dialogue-Driven Interaction and Looping Scene Control*.

Computing, Performance and Communication Systems, 10(1), 69-75. DOI: 10.23977/cpcs.2026.100108

Published case study based on *On the Other Side of the Wall*, documenting dialogue-gated progression, looping scene control, constrained traversal, and lightweight modular Unity scripting as transferable practice knowledge.

ACADEMIC & PROFESSIONAL EXPERIENCE

Lecturer in Digital Media Technology

Chongqing College of International Business and Economics | 2022-2025

- Designed and taught undergraduate courses in Interactive Technology, Virtual Reality, game-engine prototyping, and digital media production.
- Established the college's first Digital Media Studio and developed its first interaction-focused curriculum, including course frameworks, assignments, assessment structures, and project modules.
- Supervised 20+ undergraduate theses and graduation projects, guided work in games, VR, physical computing, and media installation, and organised two teaching exhibitions per academic year; recognised as an Excellent Thesis Advisor.

Executive Director

Xu Beihong Art Museum, Chongqing | 2020-2021

- Contributed to institutional setup, exhibition planning, public programmes, partner coordination, and operational development.

Multimedia Designer

Chongqing Daoshi Cultural Consulting Co., Ltd. | 2017-2019

- Designed and delivered large-scale multimedia installations and immersive environments with interdisciplinary production teams.

SELECTED RESEARCH & ARTISTIC PROJECTS

On the Other Side of the Wall

2022–2025 | Unity, C# | First-person experimental walking simulation | Published case study 2026

Links domestic, educational, workplace, and transit spaces through constrained traversal, dialogue-gated progression, looping scene control, fade-and-teleport transitions, and embedded moving-image media. The work treats movement, delay, thresholds, and system state as compositional material, and underpins my 2026 publication on reusable interaction structures in independent Unity practice.

SOMETHING BIG IS COMING

2025 | Unity | Multi-screen meta-media environment

A state-based audiovisual work built around anticipation rather than plot resolution. Television, dialogue, surveillance imagery, timed interruptions, and callable media pools are coordinated through artist-authored rules, making interface, timing, and screen relations part of the work's narrative structure. It serves as a methodological prototype for my research on audiovisual interface orchestration and bounded generative media.

Interactive Design for Deaf and Hard-of-Hearing Audiences

2024–2025 | Unity 2022.3 LTS, C# | Audio-reactive research prototype / WebGL

Explores accessibility as a starting point for audiovisual composition rather than a later addition. Real-time analysis connects musical change to movement, light, colour, materials, and environmental behaviour, while participant input alters the source audio and reorganises the visual world. The current prototype is exploratory and frames future co-design and structured evaluation with Deaf and hard-of-hearing participants.

In 2001, I Came to the Internet for the First Time

2023 | Unity, C# | Real-time interactive painting

Reconstructs early-internet visual memory through a yellow smiley, sound-reactive digital rain, a procedurally generated Windows-like landscape, and live camera imagery. The project treats the engine, shader, sound input, and camera as painting materials, using rules and repetition to examine cultural symbols, user control, and spectatorship.

You n Me

2021 / reconstructed 2026 | Browser-based networked work | p5.js, JavaScript, WebRTC / network media

A web sculpture in which live camera images from connected participants become ring-like strips wrapped around a shared 3D form. The sculpture grows with collective presence while its underlying geometry remains fixed. Its 2026 reconstruction extends the work into a study of digital preservation and the material infrastructures on which browser-based art depends.

Children's System: Eggs Eggs Eggs

2020 | Participatory multimedia installation | 3D printing, Max/MSP, sensors, responsive light

Six-month collaborative project with primary-school children developed for the O'Kids International Children's Art Festival at Yuan Art Museum. Field recordings, child-authored sculptural forms, found objects, rule-based sound, and responsive light were assembled into an altar-like public environment, linking participatory pedagogy with embodied interaction and symbolic invention.

ADDITIONAL PRACTICE

Result Fairy (2021) — interactive installation / AR / programmed ritual exploring computational folk belief and technical systems as structures of uncertainty and decision. **Soul Return (2024)** — collaborative cyber-ritual combining performance, projected media, ritual objects, and TouchDesigner-based visualisation.

TEACHING, CURRICULUM & SUPERVISION

- Practice-based teaching model connecting Unity, VR, Processing / p5.js, Arduino, sensors, real-time image systems, and physical materials with concept development, critique, documentation, and exhibition-oriented production.
- Built project sequences that move students from low-threshold technical exercises to complete interactive works, emphasising the relationship among concept, interaction, material, and presentation rather than technical polish alone.
- Supervised projects spanning games, VR environments, interactive software, creative coding, physical computing, and media installations; organised public teaching exhibitions and thesis supervision.

SELECTED EXHIBITIONS

- Taixu Huanjing — Kumamoto Museum of Art, Kumamoto, Japan, 2022.
- Result Fairy — NYU Art Gallery, New York, USA, 2021.
- Temporary Radio Station: Children Don't Lie — Yuan Art Museum, Chongqing, China, 2021.
- Children's System: Eggs Eggs Eggs — Yuan Art Museum, Chongqing, China, 2020.

MEMBERSHIPS, CERTIFICATION & RECOGNITION

- Unity Certified Educator.
- Member, China Society of Image and Graphics.
- Excellent Thesis Advisor, Chongqing College of International Business and Economics.

TECHNICAL & ARTISTIC PRACTICE

Real-time & Interactive Systems	Unity, C#, TouchDesigner, Processing / p5.js, vvvv Gamma, Arduino, sensors, camera input, OSC/MIDI-oriented workflows.
Audiovisual & Sound	Ableton Live, Logic Pro, Max/MSP, Pure Data, electronic music production, audio-reactive and sound-to-image systems.
3D / Moving Image / Visual	Blender, Cinema 4D, After Effects, Final Cut Pro, shaders / HLSL, real-time image processing, generative visuals.
Web & Networked Media	JavaScript, HTML/CSS, p5.js, browser camera APIs, WebRTC-oriented networked interaction and browser-based art.

LANGUAGES

Chinese — native | **English** — professional working proficiency | **Italian** — working knowledge

SELECTED LINKS

Portfolio & research: <https://mahanran.com>

Published article: <https://doi.org/10.23977/cpcs.2026.100108>

Project archive: <https://www.mahanran.com/works>